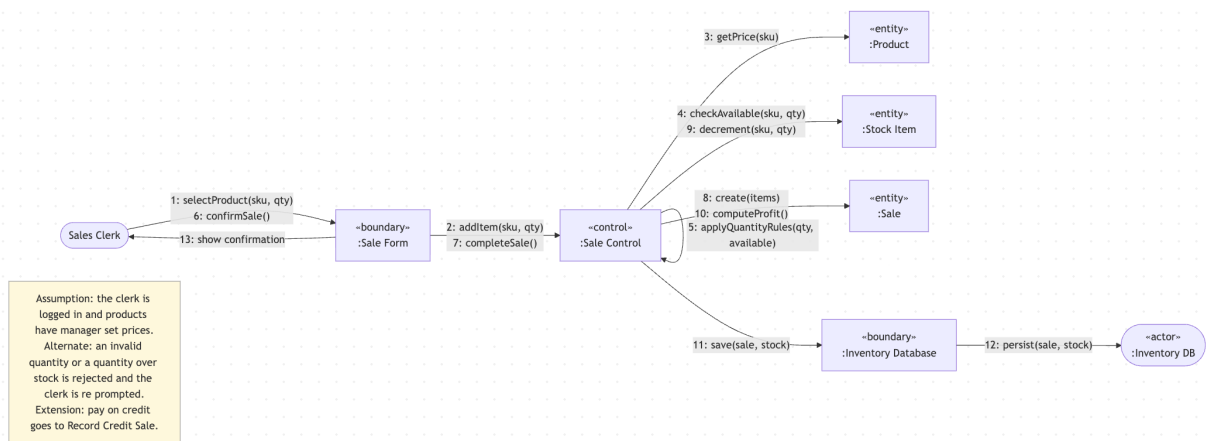


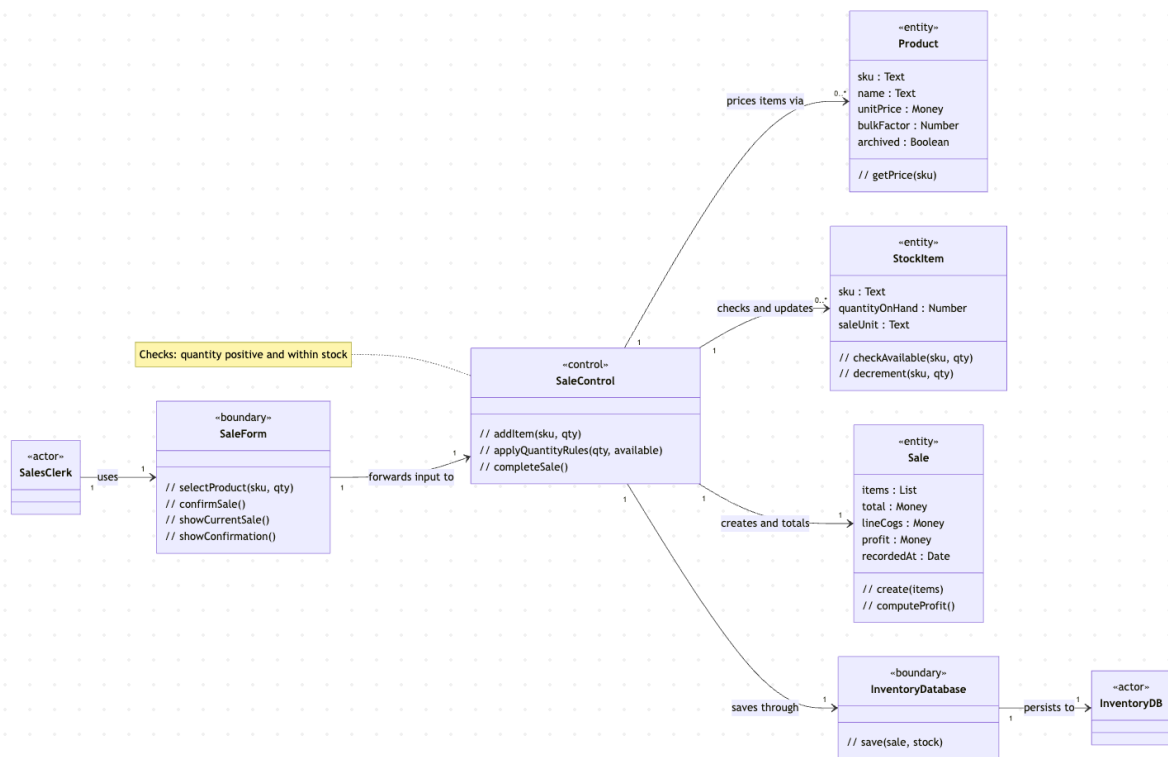
# Lab 5: Collaboration and VOPC Diagrams

## Record Sale

### Collaboration diagram



### VOPC



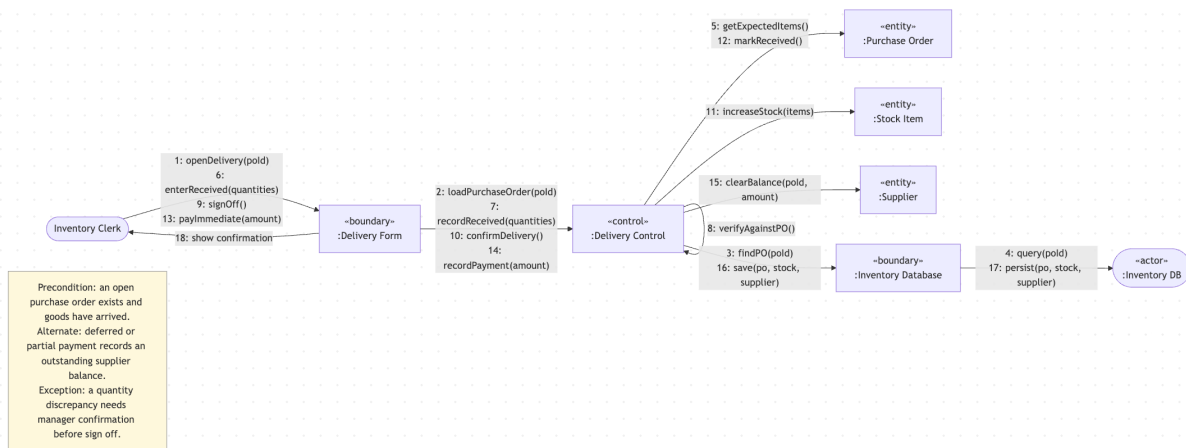
### Business rules covered

- Quantity must be positive and within available stock; the control checks this and the alternate flow rejects the line and re prompts the clerk.
- Prices are manager set; the clerk never enters a price, the control reads it from Product.
- Money rounds once at the line level (line\_cogs), so computeProfit introduces no second rounding.

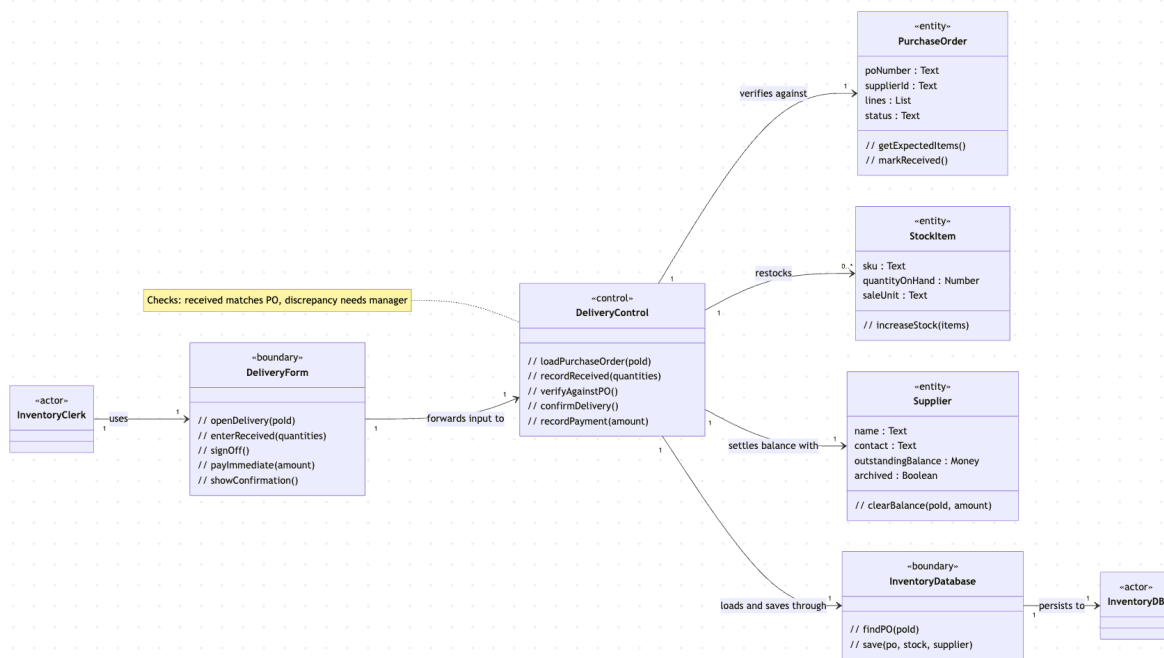
- Stock is stored in sale units, so decrement needs no unit conversion.
- Pay on credit leaves this realization through the Record Credit Sale extension.

## Record and Sign off Delivery

### Collaboration diagram



### VOPC

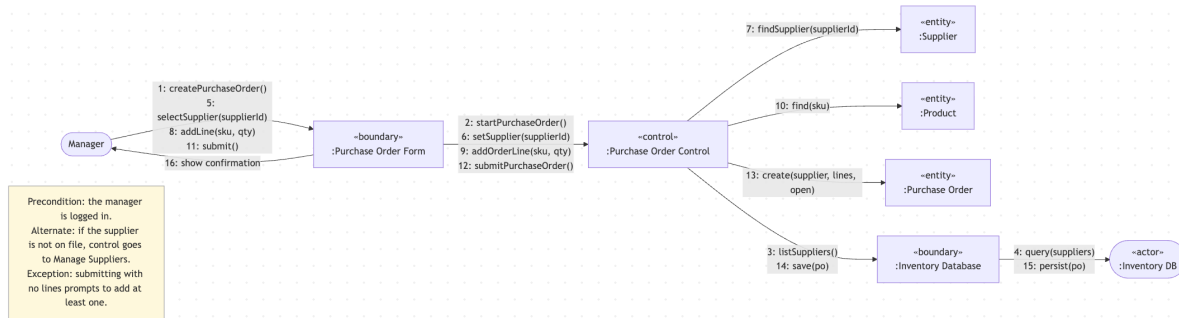


### Business rules covered

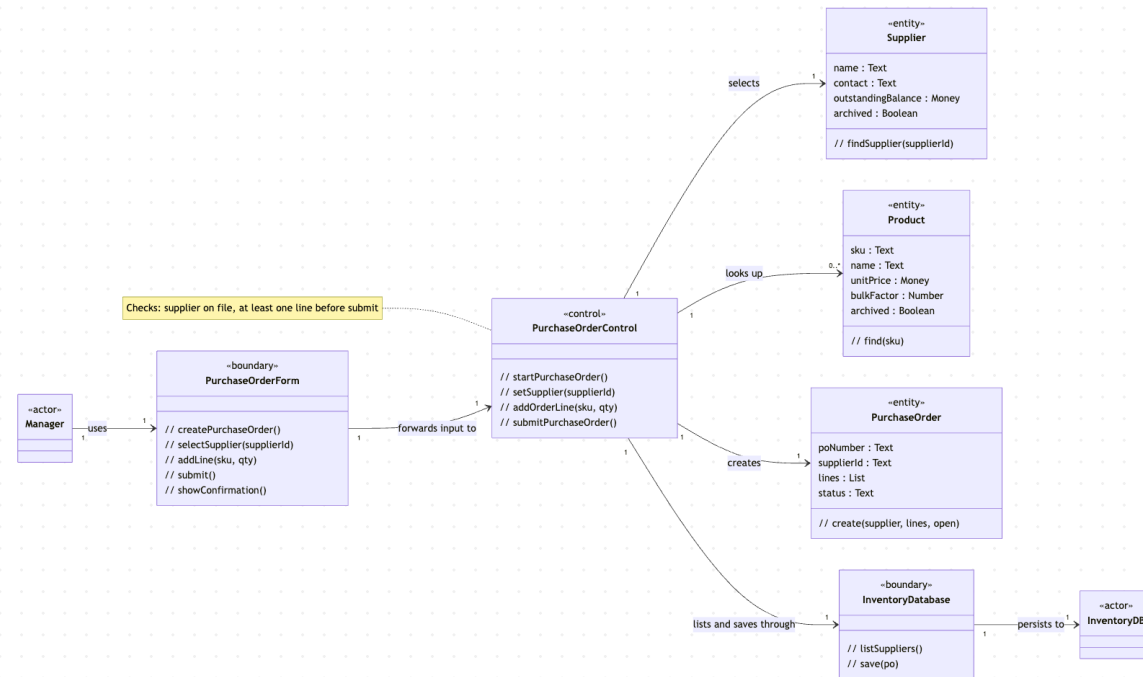
- Received quantities are verified against the purchase order; the exception path routes a discrepancy to manager confirmation before sign off.
- Supplier payment at delivery is recorded by the receiver; immediate, deferred, and partial are all allowed, and the alternate flow records an outstanding balance for deferred or partial payment.
- Standalone later payments are Manager only and are outside this realization.
- Stock increases happen only at sign off, never at data entry, so an abandoned delivery leaves stock untouched.

# Create Purchase Order

## Collaboration diagram



## VOPC



## Business rules covered

- Creating a purchase order is a Manager operation; clerks never reach this realization.
- The supplier must be on file; the alternate flow transfers control to Manage Suppliers when it is not.
- A purchase order needs at least one line; the exception at submit prompts the manager to add one.
- Suppliers and products are archived, never deleted, so supplier and product lookups exclude archived records.